

ZUZANNA DEDYK

Curriculum Vitae

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EDUCATION & EMPLOYMENT HISTORY

PhD in Neuroscience

2024 - Present

Cognitions Humaine et Artificielle Lab, École Pratique des Hautes Études - PSL, Paris, France

Systems Neuroscience Laboratory, Champalimaud Research, Champalimaud Foundation, Lisbon, Portugal (based)

Food Cognition Research Group, Institut Lyfe Research Center, Ecully, France (funding)

Thesis: "The Art and Analytics of Wine Tasting."

Supervisors: Dr. Anne-Lise Saive, Dr. Jérémie Lafraire; Collaboration: Dr. Zachary Mainen

Research Technician (Systems Neuroscience Laboratory)

2023 - 2024

Champalimaud Research, Champalimaud Foundation, Lisbon, Portugal

Latent States Camp Project: Development of cross-modal sensory experiences in flavour experiments under the supervision of Dr. Razvan Sandru and Dr. Zachary Mainen.

"WineXP" Project: Development of the project foundations under the supervision of Dr. Zachary Mainen and Dr. Anne-Lise Saive from the Institut Lyfe Research Center.

MRes in Cognitive Neuroscience (Perception, Action and Decision Making) & Minor in Neurophilosophy

2021 - 2023

Radboud University, Nijmegen, the Netherlands

Donders Institute for Brain, Behaviour and Cognition

Thesis: "Can we modulate coffee flavour perception with non-invasive brain stimulation? Probing the effect of tDCS over the IPS on flavour perception."

Supervisor: Dr. Nils Kohn

BSc in Mathematics and Natural Sciences (Neurobiology Specialisation)

2017 - 2020

Jagiellonian University, Kraków, Poland

Institute of Zoology and Biomedical Research

Thesis: "Monomolecular odour recognition - effect of training on olfactory performance."

Supervisor: Dr. Zbigniew Soltys

PROFESSIONAL EXPERIENCE

Honours Academy Research Intern (Human-Robot-Interaction Laboratory)

May 2023 – September 2023

Scuola Superiore Sant'Anna, Pisa, Italy

BioRobotics Institute

Working on evaluation of hand function transporting fragile objects: the Virtual Eggs Test (VET) in post-stroke patients in collaboration with the IRCCS Don Gnocchi Hospital in Florence under supervision of Prof. Marco Controzzi.

Research Assistant

2021 - 2023

Radboud University, Nijmegen, the Netherlands

Donders Centre for Cognition, Donders Institute for Brain, Cognition and Behaviour

Certified MRI User: Performed behavioral and fMRI data acquisition for the Healthy Brain Study and the SENSE & AI group.

Neuromodulation: Assisted in the Cognitive Neuromodulation Lab investigating the technical feasibility of combining TMS and TUS (Transcranial Ultrasound Stimulation).

Designing behavioural online experiment, implementing it on the online platform, and running a pilot for Artificial Cognitive Systems group

Mentorship: Trained new research assistants in fMRI data collection protocols.

Research Assistant (Nature Protection, Wildlife Management and Environmental Education Laboratory)

2020 - 2021

Jagiellonian University, Kraków, Poland

Institute of Environmental Sciences

Investigating the perception of single-use plastic among baristas and customers of speciality coffee shops in the era of the COVID-19 pandemic under supervision of Prof. Małgorzata Grodzińska-Jurczak. Research supported by a Student BioS Minigrant.

Summer Research Intern (Olfactory Laboratory)

July 2019 - September 2019

King's College London, United Kingdom

Centre of Developmental Neuroscience

An independent summer research project focused on associative learning in *Drosophila Melanogaster* under supervision of Dr. Camilla Larsen. Thorough introduction to confocal imaging, brain dissections, and immunostaining.

Research Assistant (Consciousness Laboratory)

January 2019 - June 2019

Jagiellonian University, Kraków, Poland

Institute of Psychology, University Hospital

Assisting with Transcranial Magnetic Stimulation (TMS) based study on the effects of TMS-induced motor activation on visual awareness.

PUBLICATIONS

Reviewed Articles

Grodzińska-Jurczak, M., Krawczyk, A., Akhshik, A., Dedyk, Z., Strzelecka, M. (2022). Contradictory or complementary? Stakeholders' perceptions of a circular economy for single-use plastics. *Waste Management*, 142, 1–8. <https://doi.org/10.1016/j.wasman.2022.01.036>

Conference Proceedings

Dedyk Z., Sandru R., Wiehe M., Mainen Z., “Exploring Cross-Modal Sensory Experiences in Flavour Perception.” Poster presented at: 30th ASIC Conference on Coffee Science (ASIC2025), Lisbon, Portugal.

Dedyk Z., Sandru R., Wiehe M., Mainen Z., “Exploring Cross-Modal Sensory Experiences in Flavour Perception.” Poster presented at: CR Symposium 2025 - “Neurocybernetics at scale”, Lisbon, Portugal.

Dedyk Z., Wiehe M., Kohn N., “Can We Modulate Coffee Flavour Perception with Non-Invasive Brain Stimulation? Probing the Effect of tDCS over the IPS on Flavour Perception.” Poster presented at: NeurotechEU Autumn School on Closed-Loop Neurotechnologies 2023, Lille, France.

Dedyk Z., Wiehe M., Kohn N., “Can we modulate coffee flavour perception with non-invasive brain stimulation? Probing the effect of tDCS over the IPS on flavour perception” Poster presented at: ECRO Conference 2023, Nijmegen, the Netherlands.

Dedyk Z., Wiehe M., Kohn N., “Can we modulate coffee flavour perception with non-invasive brain stimulation? Probing the effect of tDCS over the IPS on flavour perception” Poster presented at: ECNP Conference 2023, Barcelona, Spain.

Dedyk Z., Mikos A., Matuszyk J., Eyzaguirre Rico L., Tolosa Arias D., Grodzińska-Jurczak M., “Perception of single-use plastic among baristas and speciality coffee shops customers during the COVID-19 pandemic: A comparative study between Poland, Peru, and the UK” Oral presentation delivered at: ESAC Conference 2021, Edmonton, Canada (Remote).

Dedyk Z., Mikos A., Matuszyk J., Eyzaguirre Rico L., Tolosa Arias D., Grodzińska-Jurczak M., “How do Poles drink coffee in the pandemic era? Perception of single-use plastic among customers of specialty coffee shops” Poster presented at: CitSci Virtual – CSA Congress 2021, Arizona, US (Remote).

Dedyk Z., Sołtys Z., “Monomolecular odour recognition and the effect of expertise on olfactory performance” Poster presented at: NEURONUS 2020 IBRO Neuroscience Forum, Kraków, Poland (Remote).

Dedyk Z., Sołtys Z., “Monomolecular odour recognition and the effect of expertise on olfactory performance” Poster presented at: SymBioSe 2020 Student Conference, Groningen, Netherlands (Remote).

INVITED TALKS & SEMINARS

Institut Lyfe Research and Innovation Center, Ecully, France (2024)

Dedyk Z., Croijmans I., “Sensory Symphony: Exploring the Language of Flavors and Aromas.” Invited Research Seminar and Workshop on the language, cognition, and persuasion of coffee and perfume.

GDR O3 French Olfaction Symposium, Lyon, France (2025)

Invited Workshop Lead: “L’atelier du café.” Delivered as part of the annual national symposium on Odors, Olfaction, and Olfactometry.

AWARDS, FELLOWSHIPS & GRANTS

Doctoral Fellowship

Proposal Title: Deepfooding: Modéliser les expériences de dégustation grâce à l'intelligence artificielle
Funding Agency: Fondation Institut Lyfe
Coverage Period: October 2024 - October 2028

Honours Academy Fellowship: Beyond the Frontiers

Proposal Title: Evaluation of hand function transporting fragile objects: the Virtual Eggs Test (VET)
Funding Agency: Radboud University Honours Academy
Coverage Period: April 2023 - September 2023

Internal Grants: Student Research Minigrant

Proposal Title: Perception of single-use plastic among baristas and speciality coffee shops customers during the COVID-19 pandemic.
Funding Agency: Jagiellonian University Strategic Program Excellence Initiative (BioS Priority Research Area)
Role Status: Co-author
Total direct costs and annual budget: 7 782,25 PLN
Coverage Period: December 2020 - June 2021

Scholarships

Jeremi Króliczewski Education Trust Scholarship 2021: Awarded for a Summer School at Cambridge University
Erasmus Internship Grants: Awarded for research at the Donders Institute (2021) & King's College London (2019)

Award: Best 2nd Year Student Project SMP UJ

2nd place for the Research Project: “The effect of an affective cognitive bias on a coffee flavour perception” (2019)

PROFESSIONAL & TRADE PUBLICATIONS

Dedyk, Z. (2019) “Știință și Cafea: Percepție”, *Coffee Break Magazine*, Issue 40, pp. [52-55]. Full article available at coffee-break.ro

TEACHING

Human Olfaction Summer School, University of Dresden Medical School, Germany (2025)

Workshop Lead: "Coffee Sensory Workshop."

Institut Lyfe Research and Innovation Center, Ecully, France (2025)

Workshop Lead: "Unlocking Flavour Perception: A Coffee Sensory Neuroscience Workshop for Wine Experts."

NABA Nuova Accademia di Belle Arti, Milan, Italy (2024)

Short Course Co-Lead: "NeuroArt Flavor Fusion" (Matters of Identity week).

TECHNICAL SKILLS

Graph-Theoretical Analysis & iMDS/RSA: Quantifying cognitive maps of flavour using NetworkX (modularity, efficiency)

Programming & Analysis: Python (NumPy, NetworkX, scikit-learn), statistical modeling, mixed-effects regression

Neuroimaging & Neuromodulation (NIBS): Certified fMRI user (Siemens 3T, Donders Institute); tDCS flavor modulation (intraparietal sulcus targeting)

Sensory: Olfactometry (Sniffin' Sticks), wine/coffee sensory evaluation protocols, cross-modal psychophysics

TRAINING

Advanced Tools for Data Analysis in Neuroscience Summer School

NeuroStra, CNRS, University of Strasbourg, France (2025)

Human Olfaction Summer School (led by Prof. Thomas Hummel)

University of Dresden Medical School, Germany (2025)

Computational Neuroscience

Neuromatch Academy (Interactive Track, 2024)

Closed Loop Neurotechnologies: From Sensors to Applications Autumn School

NeurotechEU, University de Lille, France (2023)

Donders Toolkit: Non-invasive Brain Stimulation, TMS or TES

Donders Institute for Brain, Cognition and Behaviour, the Netherlands (2022)

Summer School in Materials and our environment: Striving for sustainability

University of Cambridge, United Kingdom (2021)

OUTREACH & PROFESSIONAL DEVELOPMENT

Outreach

TEDx Nova, Lisbon, Portugal (2025) Speaker Team: Responsible for speaker curation and conference organization.

Porto, Portugal (2025) Delivered a lecture titled "Factors Influencing Flavour Perception in Wine Tasting: Neurophysiology of Tasting" to support science-based training.

Bucharest Coffee Festival, Romania (2025) Workshop Lead: "Sensory Neuroscience Master Class." Led hands-on multisensory interaction sessions for coffee professionals.

Irish Brewers Cup, Ireland (2025) Workshop Lead: Sensory neuroscience training for national-level sensory judges.

Athens Coffee Festival, Greece (2024) Workshop Lead: "Coffee Talks" panel. Delivered sessions on introductory sensory neuroscience and comparative analysis of coffee and chocolate.

Polish SCA Coffee Championships, Poland (2024) Speaker: "Unlocking Flavour Perception" interactive lecture on the sensory neuroscience fundamentals for SCA Poland Educational Zone.

Moinho de Juventude, Portugal (2024) Workshop Lead: "Smell and Taste" workshop for children as part of the Science on the Walls Summer Camp.

Champalimaud Foundation, Luso, Portugal (2024) Workshop Lead: Facilitated sensory analysis in coffee for the annual research retreat.

Scuola del Caffè Ditta Artiginale, Italy (2023) Workshop Co-Lead: Ran a coffee sensory neuroscience workshop for baristas in Firenze.

Night of Biologists, Poland (2019) Workshop Lead: "Taste and Smell" workshops for middle school students.

International Brain Awareness Week, Poland (2019) Workshop Lead: "Coffee & Olfaction" at the *Neurobiological Boat* event organized by the Polish Academy of Sciences.

Service to the Community & Public

Coffee Masters Morocco, Marrakech (2025) International Judge & Trainer: Assisted in establishing the national chapter and preparing competitive standards for the Marrakech Coffee and Tea Festival.

World of Coffee (Copenhagen, 2024; Geneva, 2025) Brew Bar Manager: Orchestrated logistics and team management for international specialty coffee exhibitions.

National French Cup Tasting Championships, Paris, France (2025) Logistics & Operations: Managed technical "backend" setup for national sensory competitions.

COSYNE Conference, Lisbon, Portugal (2024) Volunteer: Provided organizational support for the Computational and Systems Neuroscience conference.

Student-led Cognitive Neuroscience Journal (2021) Senior Editor: Managed peer-review processes and editorial selection for student research.

Polish Presidential Elections, Olesno, Poland (2020) Vice-chair: Served on the polling station committee for the national electoral process.

International "Brain Bee" Contest, Kraków, Poland (2018–2020) Organizer: Coordinated semi-finals and finals of the national-level neuroscience competition for high school students.

Neuronus IBRO Neuroscience Conference (2018) Volunteer: Co-Organized the International Student conference.

Special Interests

Long-distance running Lisboa Marathon 2025 finished (4:19:20)

Sport climbing 6c+/7a